

Analyzing Ellipses: Standard Form & Graphing

Precalculus Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Find the center, vertices, and foci of an ellipse from given information
- Derive the standard equation of an ellipse using the values of a , b , and c
- Graph an ellipse by identifying key features including center, vertices, and foci

Problems

1. An ellipse has a major axis of length 10. Find the value of a .

$$2a = 10$$

2. Find the center of an ellipse whose foci are at $(1, 3)$ and $(7, 3)$.

$$\text{Center} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

3. An ellipse has foci at $(0, 0)$ and $(6, 0)$. Find the value of c , which is the distance from the center to each focus.

$$c = \frac{\text{distance between foci}}{2}$$

4. Given $a = 5$ and $c = 3$ for an ellipse, find the value of b squared using the relationship between a , b , and c .

$$c^2 = a^2 - b^2$$

5. Write the standard equation of an ellipse with center $(2, 1)$, a squared equal to 9, and b squared equal to 5, where the major axis is horizontal.

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$$



6. An ellipse has foci at (0, 1) and (4, 1) and a major axis of length 6. Find a, c, and b squared.

$$c^2 = a^2 - b^2$$

7. Find the vertices of the ellipse with center (2, 1) and a squared equal to 9, b squared equal to 5, where the major axis is horizontal.

$$\frac{(x-2)^2}{9} + \frac{(y-1)^2}{5} = 1$$

8. Given the foci at (2, 0) and (2, 4) and a major axis of length 8, write the standard equation of the ellipse. The major axis is vertical.

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$

9. Identify the center, foci, and vertices of the ellipse given by the equation below.

$$\frac{(x+3)^2}{25} + \frac{(y-4)^2}{9} = 1$$

10. An ellipse has vertices at (-1, 5) and (-1, -3) and foci at (-1, 4) and (-1, -2). Write the standard equation of the ellipse and identify b squared.

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$



Analyzing Ellipses: Standard Form & Graphing — Answer Key

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Answer Key

1. Answer: $a = 5$

- The relationship between the major axis length and a is: $2a = \text{major axis}$
- Substitute: $2a = 10$
- Divide both sides by 2: $a = 5$

2. Answer: Center = (4, 3)

- The center is the midpoint of the two foci
- x-coordinate of center: $(1 + 7) / 2 = 4$
- y-coordinate of center: $(3 + 3) / 2 = 3$
- Center = (4, 3)

3. Answer: $c = 3$

- Distance between foci = $6 - 0 = 6$
- c is half that distance: $c = 6 / 2 = 3$

4. Answer: $b^2 = 16$

- Use the formula: $c^2 = a^2 - b^2$
- Substitute: $9 = 25 - b^2$
- Solve: $b^2 = 25 - 9 = 16$

5. Answer: $(x-2)^2/9 + (y-1)^2/5 = 1$

- Center $(h, k) = (2, 1)$, $a^2 = 9$, $b^2 = 5$
- Since the major axis is horizontal, a^2 goes under $(x - h)^2$
- Equation: $(x - 2)^2 / 9 + (y - 1)^2 / 5 = 1$

6. Answer: $a = 3$, $c = 2$, $b^2 = 5$

- Major axis = 6, so $2a = 6 \rightarrow a = 3$
- Distance between foci = 4, so $c = 2$
- $b^2 = a^2 - c^2 = 9 - 4 = 5$

7. Answer: Major vertices: $(-1, 1)$ and $(5, 1)$; Minor vertices: $(2, 1+\sqrt{5})$ and $(2, 1-\sqrt{5})$

- $a = \sqrt{9} = 3$, $b = \sqrt{5} \approx 2.24$
- Major vertices (horizontal): $(2 - 3, 1) = (-1, 1)$ and $(2 + 3, 1) = (5, 1)$
- Minor vertices (vertical): $(2, 1 + \sqrt{5})$ and $(2, 1 - \sqrt{5})$

8. Answer: $(x-2)^2/12 + (y-2)^2/16 = 1$

- Center = midpoint of foci = $((2+2)/2, (0+4)/2) = (2, 2)$





- $2a = 8 \rightarrow a = 4$, so $a^2 = 16$
- $c = \text{distance from center to focus} = 2$, so $c^2 = 4$
- $b^2 = a^2 - c^2 = 16 - 4 = 12$
- Since major axis is vertical, a^2 goes under $(y - k)^2$
- Equation: $(x - 2)^2 / 12 + (y - 2)^2 / 16 = 1$

9. Answer: Center: $(-3, 4)$; Foci: $(-3 \pm 4, 4) = (1, 4)$ and $(-7, 4)$; Vertices: $(2, 4)$ and $(-8, 4)$

- Center $(h, k) = (-3, 4)$
- $a^2 = 25 \rightarrow a = 5$; $b^2 = 9 \rightarrow b = 3$
- $c^2 = a^2 - b^2 = 25 - 9 = 16 \rightarrow c = 4$
- Foci (horizontal major axis): $(-3 + 4, 4) = (1, 4)$ and $(-3 - 4, 4) = (-7, 4)$
- Major vertices: $(-3 + 5, 4) = (2, 4)$ and $(-3 - 5, 4) = (-8, 4)$

10. Answer: $(x+1)^2/7 + (y-1)^2/16 = 1$

- Center = midpoint of vertices = $((-1+(-1))/2, (5+(-3))/2) = (-1, 1)$
- $a = \text{distance from center to vertex} = |5 - 1| = 4$, so $a^2 = 16$
- $c = \text{distance from center to focus} = |4 - 1| = 3$, so $c^2 = 9$
- $b^2 = a^2 - c^2 = 16 - 9 = 7$
- Major axis is vertical, so a^2 goes under $(y - k)^2$
- Equation: $(x + 1)^2 / 7 + (y - 1)^2 / 16 = 1$

